

07/01/2025

Hiring Manager

Joby Aviation

Santa Cruz, California

Dear Hiring Manager,

I am writing to express my strong interest in the Embedded Software Engineer position at Joby Aviation company. I am confident in my ability to contribute to the development of advanced systems.

As can you see in my resume, I have worked on safety-critical systems such as the modernization of Mexico City's railway, inspection robots for fuel pipelines, and Electronic Control Units (ECUs) for the automotive industry. I believe this experience aligns well with the type of projects developed by your company, which I find both impressive and inspiring.

I have extensive experience with C and C++ programming—skills I have been developing since my university years, where embedded systems quickly became my favorite area in mechatronics engineering. I see embedded systems as an integral part of everyday life, and working with them continues to be a source of excitement and purpose for me.

Additionally, I gained hands-on experience with FPGA development during my master's degree at CIDESI, where I designed a signal acquisition architecture for Hall sensors. I also developed and implemented low-level drivers for I<sup>2</sup>C, SPI, and UART during the railway system update and glider datalogger project. Furthermore, I implemented the CAN protocol for testing and diagnostics during my time at CIDEC and BorgWarner.

My debugging experience includes identifying and resolving critical issues during the railway system update, and I also participated in the testing phase during system start-up, ensuring proper functionality in real-world conditions.

For me, using technology to improve people's lives has always been my greatest motivation. Being part of projects that contribute to accessible and efficient transportation systems is deeply fulfilling. I understand the high safety standards these systems require, having worked in industries where reliability is crucial—automotive and railways—both dedicated to human transport. For this reason, I would be honored to contribute to your aviation initiatives, which aim to provide agile and secure mobility.

I would welcome the opportunity to further discuss how my experience and skills could contribute to your team. Thank you very much for considering my application.

Sincerely,

**Ivonne Berenice Lemus Martínez**